

HATE SPEECH DETECTION USING TRANSFER LEARNING

A PROJECT REPORT

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Abstract

With the upcoming Paris 2024 Olympics, social media platforms are expected to experience a surge in global interactions, which can lead to an increase in hate speech due to varying cultural, political, and personal viewpoints. Detecting hate speech in this setting involves analyzing both text and emojis, as emojis can modify or intensify the tone of messages. Addressing this challenge is essential to maintain a positive online environment and to ensure respectful exchanges during the international event.

The primary aim of this project is to design a model that leverages both text and emoji data to improve hate speech detection accuracy. We develop modules for preprocessing text, extracting sentiment scores, encoding emojis, and analyzing context using deep learning techniques. Two models, Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM), are implemented to capture the temporal and contextual dependencies in the data. Feature extraction includes embeddings for words and emojis, N-grams, and sentiment scores, with the goal of capturing complex language patterns and diverse tones associated with hate speech in online discussions.

The system is developed on the Jupyter Notebook platform, which provides an interactive environment for data exploration, model training, and testing. Jupyter Notebook's integration with Python and compatibility with popular machine learning libraries such as TensorFlow and PyTorch make it suitable for implementing and fine-tuning complex models like LSTM and BiLSTM.